



Introduction to Robotics

Final Project

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1 Inroduction

The ABB IRB 2400/10 is one of the most widely used robotic manipulators in industry, renowned for its exceptional path accuracy and high reliability in processes such as arc welding and cutting. This project aims to conduct a complete kinematic and dynamic analysis of this six-degree-of-freedom (6-DOF) robot.

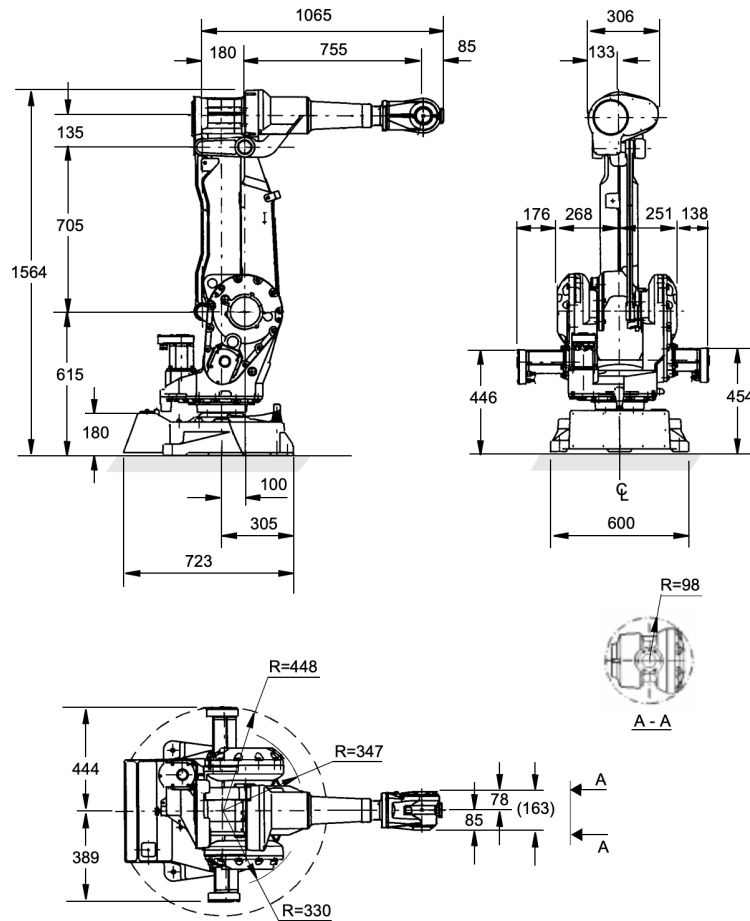


Figure 1: View of the manipulator from the side, rear and above

2 kinematic Equations of the ABB IRB 2400

2.1 Denavit-Hartenberg notation

To standardize the process, we define an outward-pointed coordinate axis. Based on this convention, we developed the D-H coordinate and the table below:

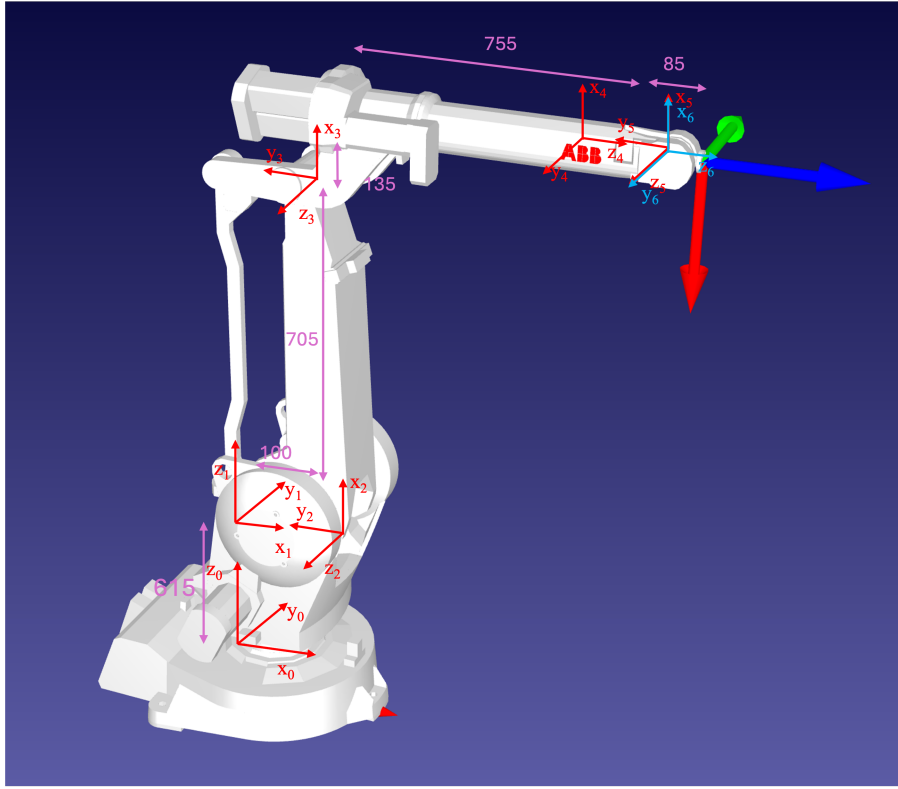


Figure 2: Link-frame assignments

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0.615	θ_1
2	-90	0.1	0	$\theta_2 + 90$
3	0	0.705	0	θ_3
4	-90	0.135	0.755	θ_4
5	90	0	0	θ_5
6	90	0	0.085	θ_6

Table 1: Link parameters of the ABB robot

2.2 Transformation Matrices

We calculate the coordinate transformation from the Tool frame T to the station frame S. The general form of ${}^{i-1}_i T$ is :

$${}^{i-1}_i T = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -d_i \sin \alpha_{i-1} \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & d_i \cos \alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

The parameters θ_i , α_{i-1} , a_{i-1} , and d_i represent the geometric characteristics of each link, as defined by the previous DH convention. Based on the general transformation equation (Eq. 1), the individual homogeneous transformation matrices were derived using the Denavit–Hartenberg parameters listed in Tab. 1. The transformations from one link frame to

the next are given by the following matrices:

$${}^0_1T = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0.615 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$${}^1_2T = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & 0.1 \\ 0 & 0 & 1 & 0 \\ -\sin \theta_2 & -\cos \theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$${}^2_3T = \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & 0.705 \\ \sin \theta_3 & \cos \theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$${}^3_4T = \begin{bmatrix} \cos \theta_4 & -\sin \theta_4 & 0 & 0.135 \\ 0 & 0 & -1 & -0.755 \\ \sin \theta_4 & \cos \theta_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

$${}^4_5T = \begin{bmatrix} \cos \theta_5 & -\sin \theta_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \sin \theta_5 & \cos \theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

$${}^5_6T = \begin{bmatrix} \cos \theta_6 & -\sin \theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0.085 \\ -\sin \theta_6 & -\cos \theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7)$$

3 Dynamics and Torque Derivation via Lagrangian Formulation

The dynamic model of the ABB IRB2400 robot is derived using the Euler-Lagrange formulation. This method relies on the energy properties of the robotic system—specifically the kinetic energy $\mathcal{K}$ and potential energy $\mathcal{P}$ —to compute the required joint torques.

The standard dynamic equation of a robotic manipulator in joint space is given by:

$$\tau = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) \quad (8)$$

where:

- $q, \dot{q}, \ddot{q} \in \mathbb{R}^6$ are the joint position, velocity, and acceleration vectors.
- $M(q) \in \mathbb{R}^{6 \times 6}$ is the symmetric, positive-definite inertia matrix.
- $C(q, \dot{q}) \dot{q} \in \mathbb{R}^6$ represents the Coriolis and centrifugal forces.
- $G(q) \in \mathbb{R}^6$ is the gravity vector.
- $\tau \in \mathbb{R}^6$ is the vector of actuator torques.

3.1 Energy Formulation

The Lagrangian $\mathcal{L}$ is defined as the difference between the total kinetic and potential energy of the system:

$$\mathcal{L}(q, \dot{q}) = \mathcal{K}(q, \dot{q}) - \mathcal{P}(q) \quad (9)$$

The equations of motion are derived using the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i, \quad i = 1, \dots, 6 \quad (10)$$

3.1.1 Kinetic Energy

The kinetic energy of the i -th link consists of both translational and rotational components:

$$\mathcal{K}_i = \frac{1}{2} m_i v_{ci}^T v_{ci} + \frac{1}{2} \omega_i^{T0} I_{ci} \omega_i \quad (11)$$

The total kinetic energy $\mathcal{K}$ is the sum of the translational and rotational kinetic energies of all six links. For a link i with mass m_i and inertia tensor I_i (defined around the center of mass), the energy is calculated as:

$$\mathcal{K} = \frac{1}{2} \dot{q}^T \left[\sum_{i=1}^6 (m_i J_{vi}^T J_{vi} + J_{\omega i}^T R_i^i I_{ci} R_i^T J_{\omega i}) \right] \dot{q} \quad (12)$$

where v_{ci} is the linear velocity of the center of mass of link i , and ω_i is the angular velocity of link i , both expressed in the base frame.

In our implementation, the relationship between kinetic energy and the inertia matrix is utilized:

$$\mathcal{K}(q, \dot{q}) = \frac{1}{2} \dot{q}^T M(q) \dot{q} \quad (13)$$

3.1.2 Potential Energy

The total potential energy $\mathcal{P}$ is determined solely by gravity acting on the link masses:

$$\mathcal{P} = \sum_{i=1}^6 m_i g^T p_{c,i} \quad (14)$$

where $g = [0, 0, 9.81]^T$ is the gravity vector and $p_{c,i}$ is the position of the center of mass of link i .

3.1.3 Mass and COM

Each link is simplified into a uniformly distributed component. The parameters were set as :

- **Link mass** m_i : [60.0, 40.0, 30.0, 20.0, 10.0, 5.0] kg
- **Center of mass position** P_{ci} : The center of mass is approximately set at the midpoint of the x -axis in the local coordinate system of each link:

```
link_com_local = np.array([[L * 0.5, 0.0, 0.0] for L in link_lengths])
```

3.2 Numerical Implementation of Dynamics

Due to the complexity of the symbolic derivation for a 6-DOF manipulator, a numerical differentiation approach (Finite Difference Method) is implemented in the simulation script to evaluate the dynamic terms in real-time.

3.2.1 Mass Matrix $M(q)$

The mass matrix $M(q)$ represents the configuration-dependent inertia of the robot. For a 6-DOF manipulator, it is a 6×6 symmetric positive-definite matrix:

$$M(q) = \begin{bmatrix} M_{11}(q) & M_{12}(q) & M_{13}(q) & M_{14}(q) & M_{15}(q) & M_{16}(q) \\ M_{21}(q) & M_{22}(q) & M_{23}(q) & M_{24}(q) & M_{25}(q) & M_{26}(q) \\ M_{31}(q) & M_{32}(q) & M_{33}(q) & M_{34}(q) & M_{35}(q) & M_{36}(q) \\ M_{41}(q) & M_{42}(q) & M_{43}(q) & M_{44}(q) & M_{45}(q) & M_{46}(q) \\ M_{51}(q) & M_{52}(q) & M_{53}(q) & M_{54}(q) & M_{55}(q) & M_{56}(q) \\ M_{61}(q) & M_{62}(q) & M_{63}(q) & M_{64}(q) & M_{65}(q) & M_{66}(q) \end{bmatrix} \quad (15)$$

Diagonal elements (M_{ii}) represent the effective inertia at joint i when only that joint is accelerating. Off-diagonal elements (M_{ij}) represent the inertial coupling between joints i and j . Due to symmetry, $M_{ij} = M_{ji}$.

In the provided script, $M(q)$ is computed using the Numerical Energy Method within the `compute_M_numeric(q)` function:

3.2.2 Rotational Inertia

To simplify calculations, the code treats the link as a slender rod, whose inertial tensor I_{ci} about its center of mass is approximately a diagonal matrix:

$$I_{ci} = \text{diag} \left(0, \frac{1}{12}m_i L_i^2, \frac{1}{12}m_i L_i^2 \right) \quad (16)$$

We use `I = (m * (L ** 2) / 12.0) * np.eye(3)` to compute the rotational inertia.

3.2.3 Gravity Vector $G(q)$

The gravity term corresponds to the gradient of the potential energy with respect to the joint angles:

$$G(q) = \frac{\partial \mathcal{P}}{\partial q} \quad (17)$$

In the simulation, the code calculates the rate of change of potential energy by applying a small displacement ϵ to each joint:

$$G_i(q) = \frac{\partial \mathcal{P}}{\partial q_i} = \frac{\partial}{\partial q_i} \sum_{j=1}^6 m_j g^T P_{cj} \quad (18)$$

This was computed by `Gq[i] = (Uplus - U) / eps` in the code

3.2.4 Coriolis and Centrifugal Terms

The Coriolis and centrifugal term $C(q, \dot{q})\dot{q}$ represents the non-linear dynamic effects. It is computed by evaluating the time derivative of the generalized momentum $p = M(q)\dot{q}$. The approximation is derived as:

$$C(q, \dot{q})\dot{q} \approx \frac{M(q + \dot{q}dt)\dot{q} - M(q)\dot{q}}{dt} \quad (19)$$

In our script, this is efficiently estimated by perturbing the system state over a small time step dt to find the rate of change of the inertial forces within `compute_Cqd_numeric(q, qd)`

3.3 Computed Torque

Finally, the required torque τ for the planned trajectory is computed at each time step by summing these components:

$$\tau_{total} = M_{num}(q)\ddot{q}_{des} + C_{num}(q, \dot{q})\dot{q}_{des} + G_{num}(q) \quad (20)$$

This allows for the analysis of the dynamic load on the ABB IRB2400 motors during the welding task.

4 Results and Discussion

This section presents the simulation results of the ABB IRB2400/10 manipulator for the complete welding task. The results include inverse kinematics solutions, trajectory planning performance, joint-level kinematic behavior, and dynamic feasibility analysis. The purpose of this section is to verify whether the planned motion satisfies the project requirements and whether the robot can safely execute the task under the given constraints.

4.1 Inverse Kinematics Results

Inverse kinematics was first solved for the given initial end-effector pose $[x, y, z, \alpha, \beta, \gamma] = [0.4, 0.4, 1.5, 0, 0, 0]$ in the station frame. A valid joint configuration was successfully obtained within a small number of iterations, indicating good numerical convergence of the solver.

The resulting joint angles are physically reasonable and lie well within the allowable joint ranges. This confirms that the target pose is reachable by the manipulator without encountering kinematic singularities or joint limit violations.

A second inverse kinematics computation was performed to obtain the joint configuration corresponding to the starting point of the welding circle. By using the previous solution as the initial guess and applying angle unwrapping, a continuous joint-space transition between the two poses was ensured. This continuity is essential for smooth trajectory planning.

4.2 Trajectory Planning from Initial Pose to Welding Start Point

A synchronized S-curve trajectory was used to plan the motion from the initial pose to the welding start point. The objective was to reach the target configuration as quickly as possible while respecting joint velocity and acceleration limits specified for the IRB2400/10.

The resulting transition time is approximately 1.26 s. During this phase, the joints accelerate and decelerate smoothly, without sharp discontinuities in velocity or acceleration. This confirms that the trajectory planning method successfully balances speed and smoothness.

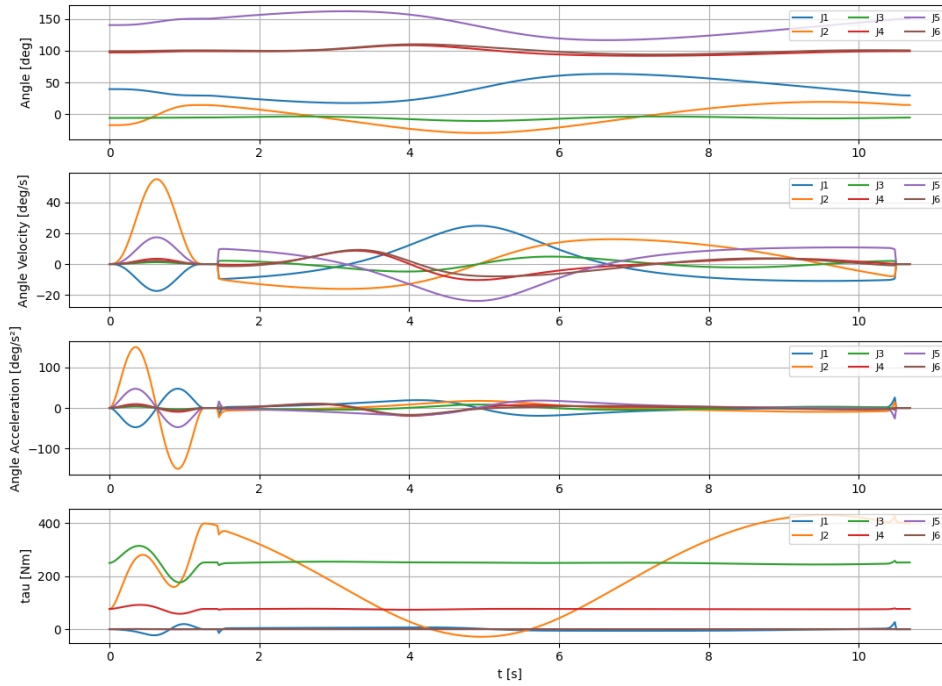


Figure 3: Joint position, velocity, acceleration, and torque profiles for the complete motion of the ABB IRB2400/10, including the transition to the welding start point and the circular welding process.

The joint position plots in Fig. 3 show smooth and continuous trajectories for all six joints during the transition phase. No joint exhibits abrupt changes or oscillatory behavior, indicating that the planned trajectory is well conditioned.

4.3 Welding Trajectory Execution

After reaching the welding start point, the end-effector follows a circular trajectory in the horizontal plane with a constant height of $z = 1.5$ m. The circular motion is executed in the clockwise direction with a prescribed tangential velocity of 0.2 m/s.

To avoid sudden changes in motion, a smooth ramp-up and ramp-down strategy is applied at the beginning and end of the welding process. As a result, the end-effector velocity transitions gradually into and out of the steady circular motion.

This behavior is reflected in the joint-space trajectories, where joint velocities become approximately periodic during the welding phase, corresponding to the repetitive nature of the circular motion.

4.4 Joint Velocity and Acceleration Analysis

The joint velocity and acceleration profiles are shown in Fig. 3. During the initial transition phase, several joints exhibit relatively higher velocity and acceleration peaks as the robot moves rapidly to the welding start point. Among all joints, Joint 2 reaches the highest velocity magnitude, which is expected due to its dominant role in positioning the arm in the horizontal plane.

Despite these peaks, all joint velocities remain significantly below their maximum allowable limits. Similarly, joint accelerations stay within the specified acceleration bounds throughout the entire motion.

Small oscillations can be observed in the acceleration profiles near the transition between the point-to-point motion and the welding motion. These effects are primarily due to numerical differentiation of discretized joint trajectories and the switching between trajectory segments, rather than indicating physical instability of the robot.

Overall, the velocity and acceleration results confirm that the planned trajectory satisfies the kinematic constraints of the manipulator.

4.5 Torque Analysis and Dynamic Feasibility

The joint torque profiles computed using an approximate Newton–Euler dynamic model are shown in the bottom subplot of Fig. 3. The results provide insight into the dynamic feasibility of the task.

Joints 2 and 3 experience the largest torque demands during the motion. This is consistent with their role in supporting the majority of the manipulator mass and maintaining the position of the end-effector during the welding process. In contrast, the wrist joints (Joints 4–6) exhibit much smaller torque magnitudes, reflecting their lower inertial and gravitational loads.

Although noticeable torque variations occur during the initial acceleration phase, all torque values remain below the estimated actuator capabilities. This indicates that the robot can execute the complete motion, including the welding trajectory, without exceeding its torque limits.

4.6 Overall Discussion

The results demonstrate that the proposed kinematic modeling, inverse kinematics solver, and trajectory planning strategy successfully enable the ABB IRB2400/10 to perform the specified welding task.

The joint motions are smooth and continuous, the velocity and acceleration constraints are respected, and the torque requirements remain within acceptable ranges. These results collectively confirm that the robot is capable of safely and reliably carrying out the circular welding operation defined in the project requirements.

5 Conclusion

In this project, a complete kinematic, trajectory planning, and dynamic analysis framework was developed for the ABB IRB2400/10 industrial manipulator to execute a circular welding task. The objective was to determine whether the robot can reach the specified poses, follow the required welding trajectory smoothly, and perform the motion within its kinematic and dynamic limits.

First, the geometric structure of the ABB IRB2400/10 was modeled using the Modified Denavit–Hartenberg (MDH) convention. Based on this representation, the MDH table was constructed and forward kinematics was implemented to compute the transformation between the station frame and the tool frame, establishing a consistent relationship between joint space and Cartesian space.

Inverse kinematics was then solved for the specified initial end-effector pose as well as for the starting point of the welding circle. Valid joint configurations were obtained for both poses, demonstrating that the target positions are reachable without encountering singularities or joint limit violations. Continuity between successive inverse kinematics solutions was enforced to ensure smooth joint-space motion.

A synchronized S-curve trajectory planning strategy was applied to move the robot from the initial pose to the welding start point as quickly as possible while respecting joint velocity and acceleration constraints. The maximum joint velocities and accelerations were identified and compared with the specified limits, and all values were found to remain well within the allowable ranges. After reaching the start point, the robot executed a circular welding trajectory in the horizontal plane with a constant height and prescribed tangential velocity, during which joint positions and velocities were continuously computed.

An approximate dynamic model based on the Newton–Euler formulation was used to calculate the joint torques throughout the entire motion. The results show that the largest torque demands occur at the shoulder and elbow joints, while the wrist joints experience relatively small torques. All computed torques remain within reasonable bounds, indicating that the robot can physically carry out the welding task.

Finally, an animation of the complete motion was generated to visually verify the correctness of the kinematic modeling, trajectory planning, and dynamic behavior.

In summary, the results presented in this report confirm that the ABB IRB2400/10 is capable of safely and reliably performing the specified circular welding operation. The methodology developed in this project provides a systematic approach for analyzing industrial robot tasks and can be extended to more complex trajectories and dynamic scenarios in future work.

6 Reference

[1] P. Obal and P. Gierlak, "A Comprehensive Model of the ABB IRB 2400 for Simulation and Control Applications," *Acta Mechanica et Automatica*, vol. 19, no. 1, 2025. DOI: 10.2478/ama-2025-0064.

[2] ABB Robotics, "Product Specification IRB 2400 M98A," ABB Library. [Online]. Available: Product Specification [Accessed: Jan. 24, 2026].